

An Explainable MaxViT Framework with Gated Residual Feature Modulation for Histopathology Classification

Abstract—Automated classification of lung and colon histopathology images can help improve cancer assessment, which is often slow and inconsistent. However, many existing studies remain subject to evaluation bias caused by duplicate data and provide limited evidence regarding calibration, robustness, and interpretability. In this study, an explainable MaxViT-Tiny framework with a Gated Residual Feature Modulation (GRFM) module is proposed to improve the classification of five lung and colon histopathological classes. Exact and near-duplicate images were removed from LC25000 before subset construction. GRFM adaptively refines the pooled transformer representation through gated feature transformation and residual preservation. On an unseen 305-image test set, the model achieved 99.67% accuracy, 99.68% macro precision, 99.68% macro recall, and 99.67% macro F1-score, with MCC and Cohen’s kappa values of 0.9959. It also yielded a Brier score of 0.0064 and an expected calibration error of 0.0032. Three-fold cross-validation produced a mean accuracy of $99.79\% \pm 0.07\%$, while multi-seed testing confirmed the stability of the results. In the ablation study, adding GRFM increased the accuracy from 97.38% to 99.67% with only a 3.46% increase in trainable parameters. Corruption testing showed good robustness to intensity changes and occlusion, whereas blur and noise remained challenging. Grad-CAM++, refinement-change maps, and saliency mapping highlighted diagnostically relevant cellular and glandular structures. These results provide a reliable, calibrated, interpretable, and computationally practical framework; however, external validation in multicenter settings is still required.

Index Terms—Lung and colon cancer; Histopathology image classification; MaxViT-Tiny; Gated Residual Feature Modulation; Duplicate-aware evaluation.

I. INTRODUCTION

Histopathology is a key step in the diagnosis of malignant tissue in the lungs and colon [1]. However, its assessment is time-consuming, relies on specialist expertise, and can be affected by inter-observer variability. Deep learning has therefore been extensively studied as a computer-aided approach for pattern recognition in digitized histopathology images. Recently, convolutional, hybrid, and attention-based models have achieved very strong results on the LC25000 dataset, reaching near-perfect or even perfect scores. Despite these advances, high benchmark accuracy does not necessarily translate into reliable or clinically meaningful behavior.

Several issues remain insufficiently addressed. First, LC25000 contains augmented and visually similar images, which may result in exact or near-duplicate samples appearing across data partitions and inflating performance estimates [2]. Second, many studies focus primarily on accuracy, precision, and recall, while providing limited analysis of calibration, statistical uncertainty, random-seed stability, or robustness to common image degradations [1]. Third, although attention and ensemble methods can improve representation learning, they may also increase architectural complexity, and the contribution of newly introduced modules is not always isolated through backbone-level ablation. Finally, studies often do not examine whether feature refinement preserves diagnostically meaningful tissue regions in post-hoc explanations.

To address these limitations, this study introduces an explainable MaxViT-Tiny framework with a Gated Residual Feature Modulation (GRFM) module for five-class lung and colon histopathology classification. Exact and near-duplicate images are removed before model

development using cryptographic, pixel-level, and perceptual hashing. MaxViT-Tiny learns both local morphological characteristics and long-range spatial dependencies, while GRFM dynamically refines the pooled transformer representation through gated transformation and residual preservation. The proposed framework is compared with compact transformer baselines and the unmodified MaxViT-Tiny backbone under a unified training protocol.

The main contributions of this work are summarized as follows:

- A duplicate-aware LC25000 preparation strategy is introduced to remove exact and near-duplicate images before data splitting, thereby reducing leakage-related evaluation bias.
- A Gated Residual Feature Modulation (GRFM) module is integrated with MaxViT-Tiny to adaptively refine the pooled transformer features while preserving the original representation through residual learning.
- A comprehensive evaluation protocol is conducted using accuracy, precision, recall, F1-score, MCC, Cohen’s kappa, sensitivity, specificity, calibration metrics, bootstrap confidence intervals, cross-validation, random-seed stability, corruption robustness, and computational efficiency.
- A multi-level explainability analysis is performed using backbone activation maps, GRFM-guided representations, refinement-change maps, Grad-CAM++, and gradient-based saliency maps to examine the regions influencing model predictions.

Overall, the proposed framework extends conventional accuracy-based assessment by jointly considering predictive discrimination, probability calibration, stability, robustness, computational practicality, and visual interpretability. The study therefore provides a more comprehensive and transparent evaluation of transformer-based lung and colon histopathology classification and demonstrates the contribution of GRFM to feature refinement and classification performance.

II. LITERATURE REVIEW

Automated diagnosis of lung cancer from CT images and classification of lung and colon histopathological images have progressed considerably with deep learning [3]. In recent years, CNNs, vision transformers, hybrid architectures, attention mechanisms, and explainable AI (XAI) have achieved significant advances [4], [5]. This section summarizes these developments and highlights unresolved issues related to reliability, robustness, calibration, and generalization that motivate the proposed approach.

A. Hybrid and Explainable Histopathology Classification

Recent histopathology studies on lung and colon cancer classification have increasingly adopted hybrid and ensemble learning approaches. Vanitha *et al.* used Xception and MobileNet with hyperparameter optimization and Grad-CAM to achieve an accuracy of 99.44% in Lung Colon XAI [6]. Alotaibi *et al.* proposed the HIELCC-EDL framework, which integrates multiple classifiers, a channel-attention ResNet50, Tuna Swarm Optimization, and Wiener filtering, and achieved an accuracy of up to 99.60% [7]. Similarly, Gowthamy

and Ramesh combined pretrained deep features with a Kernel Extreme Learning Machine (KELM) and achieved a classification accuracy of 98.9% [8]. These studies demonstrate the effectiveness of complementary feature extraction and classifier fusion; however, their evaluations mainly emphasize final benchmark accuracy. Explainability has also become an important component of histopathology classification. Hasan *et al.* proposed a lightweight multiscale CNN incorporating Grad-CAM and SHAP, which achieved 99.20% accuracy with approximately 1.1 million trainable parameters [9]. Shahadat *et al.* proposed a compact Squeeze-and-Excitation-based network that achieved 100% accuracy on the dataset [10]. These approaches improved interpretability and deployment efficiency, but their explanations were largely qualitative, with limited analysis of probability calibration, statistical uncertainty, and performance under image corruption.

B. Lightweight Multiscale and Attention-Based Models

As performance on LC25000 approached saturation, researchers increasingly explored compact multiscale and attention-based architectures. Attallah combined multiscale features extracted from compact CNNs with feature selection and SVM classifiers, achieving an accuracy of 99.3% [11]. Building on this direction, Ochoa-Ornelas proposed BiLight-Attn, which combines EfficientNetV2-B0 and MobileNetV3-Small through cross-gated attention and achieved 99.84% accuracy [12]. AlShehri developed DPCSE-Net, which integrates DenseNet121, MobileNetV3-Large, Squeeze-and-Excitation, and multi-head self-attention, achieving an accuracy of 99.88% [13]. These studies demonstrate the ability of attention-based feature fusion to capture both local cellular details and broader tissue organization. However, most approaches rely on combining complete backbone networks or applying attention modules to spatial feature maps. Comparatively little work has focused on refining the final transformer representation through a lightweight gated residual modulation mechanism before classification.

C. Generalization and Reliability-Aware Evaluation

Several attempts have been made to improve evaluation reliability beyond a single train–test split. Uddin *et al.* evaluated resilient architectures on histopathology and CT datasets and achieved 99.53% accuracy on LC25000 [14]. Ochoa-Ornelas *et al.* removed redundant LC25000 patches before combining MobileNetV2, EfficientNetB3, Grey Wolf Optimization, and machine-learning classifiers [15]. Although their reported accuracy of 94.8% was lower than many near-perfect results, the duplicate-aware evaluation provided a more conservative performance estimate. Anusha and Reddy combined handcrafted descriptors, EfficientNetB0 features, transformer attention, and incremental learning [16]. They achieved 99.87% accuracy on LC25000 and accuracies of 99.07% and 98.4% on two external datasets. Rahman *et al.* subsequently proposed a multi-head-attention CNN ensemble with ensemble Grad-CAM, achieving 100% across the major classification metrics under k-fold validation [17]. These findings demonstrate the importance of cross-validation, external testing, and explainability. However, evaluation practices remain inconsistent, and many studies do not jointly report calibration errors, confidence intervals, random-seed stability, corruption robustness, computational requirements, and component-level ablation.

D. Research Gap and Study Positioning

To address the limitations discussed above, this study enhances MaxViT-Tiny with GRFM to develop a framework for five-class lung

and colon histopathology classification. Unlike many previous studies, exact and near-duplicate images are removed from the dataset before splitting to reduce evaluation bias. The proposed Gated Residual Feature Modulation (GRFM) module refines the pooled transformer representation with minimal computational overhead and is evaluated through backbone-level ablation against the original MaxViT-Tiny. In addition to classification accuracy, the proposed framework is evaluated using the Matthews Correlation Coefficient (MCC), Cohen’s kappa, sensitivity, specificity, Brier score, Expected Calibration Error (ECE), bootstrap confidence intervals, cross-validation, random-seed stability, corruption robustness, computational profiling, and Grad-CAM++ visualization. This comprehensive assessment considers the model’s reliability, robustness, interpretability, computational practicality, and predictive performance.

III. METHODOLOGY

The proposed methodology was designed as a reliable, interpretable, and computationally efficient transformer-based framework for multiclass lung and colon histopathology image classification. The pipeline as shown in fig 1 comprises several stages: LC25000 image collection, duplicate removal, class-based subset construction, image preprocessing, dataset splitting, training-set augmentation, transformer-based feature learning, GRFM-based feature refinement, model optimization, and final evaluation. These stages were designed to minimize data leakage, preserve the diagnostic value of tissue patterns, maximize training diversity, and facilitate a fair comparison among the five histopathological categories.

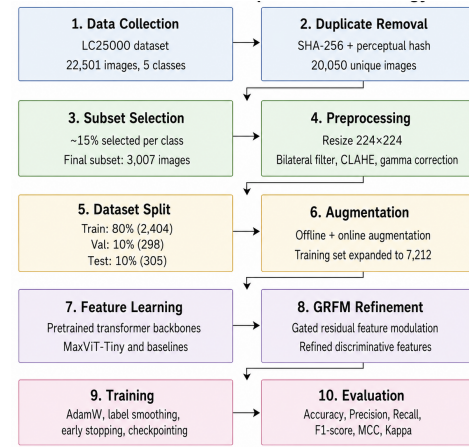


Fig. 1. Workflow of the proposed methodology for lung and colon histopathology image classification.

A. Data Collection

This study used the LC25000 lung and colon histopathology image dataset obtained from Kaggle. The dataset contained 22,501 images distributed across five classes: colon adenocarcinoma, normal colon, lung adenocarcinoma, normal lung, and lung squamous cell carcinoma. Three classes represent malignant tissue patterns, whereas the other two represent non-malignant colon and lung tissue patterns. The dataset contains variations in tissue organization, nuclear appearance, glandular structure, texture, and staining intensity, making it suitable for evaluating deep learning-based histopathology classification. The original, cleaned, and selected subset distributions are summarized in Table I.

TABLE I
CLASS-WISE DESCRIPTION AND DISTRIBUTION OF THE LC25000
DATASET.

Class	Description	Original	Cleaned	Selected
colon_aca	Colon tissue showing adenocarcinoma-related histopathological patterns.	4500	4009	601
colon_n	Non-malignant colon tissue with comparatively normal tissue organization.	4500	4060	609
lung_aca	Lung tissue showing adenocarcinoma-related cellular and structural patterns.	4500	3969	595
lung_n	Non-malignant lung tissue without the malignant patterns represented by the cancer classes.	4500	3952	593
lung_scc	Lung tissue showing squamous cell carcinoma-related histopathological patterns.	4501	4060	609
Total	—	22501	20050	3007

B. Duplicate Detection and Subset Construction

Duplicate images were removed before dataset splitting to prevent identical or highly similar images from appearing in multiple partitions. Exact duplicates were identified using SHA-256 hashes of both the original files and decoded pixel data, whereas near-duplicates were detected using a 64-bit perceptual hash with a maximum Hamming distance of 5. This process removed 1,010 exact duplicates and 1,441 near-duplicates, leaving 20,050 unique images from the original 22,501 images. Finally, to reduce computational requirements while preserving the class distribution, approximately 15% of the cleaned images were randomly selected from each class using a fixed seed of 42.

C. Data Preprocessing

A preprocessing strategy was adopted to provide the model with a consistent input representation while preserving diagnostically relevant tissue structures and staining characteristics in the selected histopathology images. Each image was resized to 224×224 pixels using area interpolation to satisfy the input-size requirements of the evaluated transformer models. Mild edge-preserving denoising was then applied using a bilateral filter with $d = 5$, $\sigma_{\text{Color}} = 25$, and $\sigma_{\text{Space}} = 25$. Contrast was enhanced by applying CLAHE to the luminance channel of the CIELAB color space with a clip limit of 1.8 and a tile-grid size of 8×8 . This procedure improved local contrast while limiting undesirable changes to the original stain colors. Finally, gamma correction with $\gamma = 1.05$ was applied. All 3,007 processed images were clipped to the valid intensity range and saved as 8-bit PNG files in their respective class folders.

D. Dataset Splitting and Training-Set Augmentation

After preprocessing, the 3,007 images were split class-wise into training, validation, and test sets using a ratio of 0.8 : 0.1 : 0.1, as presented in Table II, with a fixed random seed of 42. This procedure produced 2,404 training images, 298 validation images, and 305 unseen test images. The validation set was used for early stopping, checkpoint selection, and learning-rate adjustment during training. Offline augmentation was applied only to the training set, with two augmented images generated from each original image using rotation, translation, zooming, horizontal flipping, and nearest-neighbor boundary filling. Including the original images, this process produced 7,212 training images. During data loading, additional online transformations, including horizontal and vertical flipping, rotation, color jittering, and random erasing, were applied to improve

model generalization. All images were converted to tensors and normalized using the ImageNet mean and standard deviation. No random augmentation was applied to the validation or test sets.

TABLE II
CLASS-WISE DISTRIBUTION AFTER DATASET SPLITTING AND OFFLINE
AUGMENTATION.

Class	Original Train	Augmented Train	Validation	Test
Colon adenocarcinoma	480	1440	60	61
Normal colon	487	1461	60	62
Lung adenocarcinoma	476	1428	59	60
Normal lung	474	1422	59	60
Lung squamous cell carcinoma	487	1461	60	62
Total	2404	7212	298	305

E. Proposed GRFM-Enhanced MaxViT-Tiny Architecture

The proposed model integrates an ImageNet-pretrained MaxViT-Tiny backbone with a Gated Residual Feature Modulation (GRFM) module for multiclass lung and colon histopathology classification. MaxViT-Tiny captures local tissue characteristics and long-range spatial relationships through convolutional operations, block attention, and grid attention. For an input image I , the globally pooled feature representation generated by the backbone is expressed as

$$\mathbf{x} = F_{\text{MaxViT}}(I),$$

where $F_{\text{MaxViT}}(\cdot)$ denotes the MaxViT-Tiny feature-extraction function.

The extracted feature vector is projected into a 512-dimensional representation and passed through separate feature-transformation and sigmoid-gating branches. The gating branch adaptively emphasizes informative histopathological features, while the residual pathway preserves the original projected representation and supports stable information flow. The GRFM-refined representation is defined as

$$\mathbf{f} = \text{Dropout}(\text{LN}(\mathbf{p} + \mathbf{t} \odot \mathbf{g})),$$

where $\mathbf{p}$, $\mathbf{t}$, and $\mathbf{g}$ denote the projected features, transformed features, and modulation gate, respectively, and $\odot$ represents element-wise multiplication. The refined representation is subsequently processed by a classification head composed of a linear layer, layer normalization, GELU activation, dropout, and a final five-class output layer. The complete architecture and information flow of the proposed GRFM-enhanced MaxViT-Tiny model are illustrated in Fig. 2.

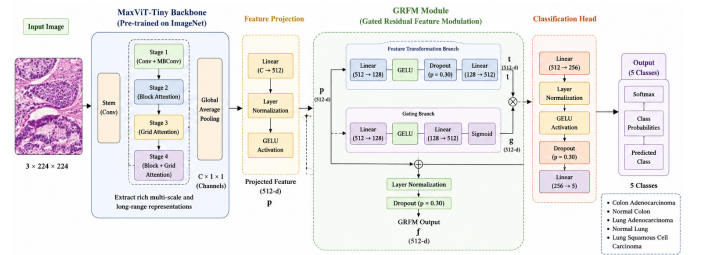


Fig. 2. Architecture of the proposed GRFM-enhanced MaxViT-Tiny model.

F. Baseline Models and Ablation Study

Four compact transformer-based classifiers-MobileViT-S, Swin-Tiny, DeiT-Tiny, and TinyViT-were selected for comparison with the proposed MaxViT-Tiny-GRFM model. All baseline models were initialized with ImageNet-pretrained weights and trained using the same data splits, training protocol, and evaluation settings. To assess

TABLE III
UNIFIED TRAINING AND OPTIMIZATION SETTINGS USED FOR ALL MODELS.

Setting	Configuration
Input size	224 × 224 pixels
Number of classes	5
Weight initialization	ImageNet pretrained weights
Fine-tuning strategy	Full-network fine-tuning
Maximum epochs	35
Batch size	16
Data-loader workers	2
Optimizer	AdamW
Learning rate	2×10^{-5}
Weight decay	1×10^{-4}
Loss function	Cross-entropy with 0.10 label smoothing
Gradient clipping	Maximum norm = 1.0
Scheduler	ReduceLROnPlateau
Scheduler factor	0.5
Scheduler patience	2 epochs
Minimum learning rate	1×10^{-7}
Early stopping patience	7 epochs
Mixed precision	Enabled (CUDA)
Checkpoint criterion	Best validation macro F1-score

the contribution of gated feature modulation and residual refinement, an ablation study was also conducted by comparing the proposed MaxViT-Tiny-GRFM model with the original MaxViT-Tiny backbone without the GRFM module.

G. Model Training Settings

All models were trained under identical optimization settings to ensure a fair comparison. Training was performed in a CUDA-enabled Kaggle Notebook using deterministic random seeds, mixed-precision computation, and full-network fine-tuning. The common training configuration is summarized in Table III.

H. Model Explainability Using Grad-CAM++

Grad-CAM++ was used to visually explain the predictions of the proposed MaxViT-Tiny-GRFM model by identifying the spatial regions that contributed most strongly to each predicted histopathological class. Activation maps were extracted from the final spatial feature stage of the MaxViT-Tiny backbone before global average pooling and were then mapped onto the corresponding tissue images. These visualizations help determine whether the model focuses on diagnostically relevant regions, such as abnormal glandular structures, densely cellular areas, nuclear alterations, and tissue boundaries, rather than on unrelated image regions.

IV. RESULTS AND DISCUSSION

A. Comparative Test Performance Analysis

To evaluate classification performance, the proposed MaxViT-Tiny-GRFM model was tested on an unseen set of 305 histopathology images and compared with the Swin-Tiny, DeiT-Tiny, and TinyViT models. A summary of the macro-averaged performance metrics obtained on the unseen test set is provided in Table IV.

Among the compared methods, the proposed MaxViT-Tiny-GRFM model achieved the highest accuracy of 0.9967, macro precision of 0.9968, macro recall of 0.9968, and macro F1-score of 0.9967. The model correctly classified 304 of the 305 test images, demonstrating strong generalization performance.

TABLE IV
COMPARATIVE CLASSIFICATION PERFORMANCE ON THE UNSEEN TEST SET.

Model	Acc.	Prec.	Rec.	F1
Swin-Tiny	0.8131	0.8123	0.8131	0.8111
DeiT-Tiny	0.9311	0.9347	0.9310	0.9318
TinyViT	0.9770	0.9791	0.9768	0.9770
Proposed MaxViT-Tiny-GRFM	0.9967	0.9968	0.9968	0.9967

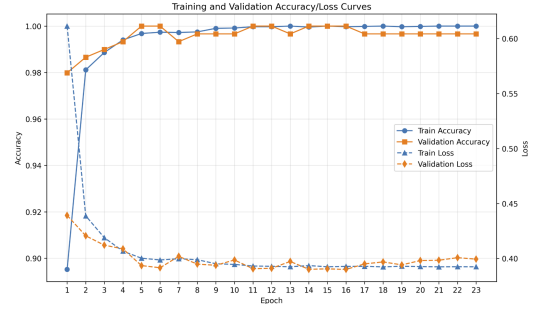


Fig. 3. Accuracy and Loss of the proposed MaxViT-Tiny-GRFM model.

Among the baseline models, TinyViT achieved the second-highest accuracy and macro F1-score, both of which were 0.9770. DeiT-Tiny achieved an accuracy of 0.9311 and a macro F1-score of 0.9318, whereas Swin-Tiny showed the lowest performance, with an accuracy of 0.8131 and a macro F1-score of 0.8111. Overall, the proposed MaxViT-Tiny-GRFM model outperformed all reported transformer baselines across the evaluation metrics. The proposed framework was also assessed through calibration analysis, confidence-interval estimation, ablation studies, cross-validation, random-seed stability analysis, corruption robustness testing, computational profiling, feature-space visualization, and Grad-CAM++ explanations, thereby extending the evaluation beyond test-set accuracy.

B. Training Convergence Analysis

The proposed MaxViT-Tiny-GRFM model converged rapidly during the initial training epochs and reached a validation accuracy of 100% at epoch 5. The optimal checkpoint was selected at epoch 16, at which the lowest validation loss of 0.3900 was obtained together with perfect validation accuracy and macro F1-score. The learning-rate scheduler gradually reduced the learning rate after the validation performance began to plateau, and training was stopped at epoch 23 through early stopping. The training and validation accuracy curves are presented in Figure 3. The curves indicate stable convergence and only a small gap between training and validation accuracy, with no evident overfitting on the internal validation set.

C. Reliability, Calibration, and Statistical Validation

Table V presents the reliability and calibration results of the proposed MaxViT-Tiny-GRFM model. The MCC and Cohen's kappa values of 0.9959 and 0.9971, respectively, demonstrate near-perfect agreement between the model predictions and the ground-truth labels. The model also achieved a high macro sensitivity of 0.9968 and specificity of 0.9992. Its low Brier score of 0.0064 and Expected Calibration Error (ECE) of 0.0032 indicate well-calibrated probability estimates. Moreover, the narrow 95% confidence intervals for accuracy and macro F1-score indicate reliable and stable performance on the internal test set.

TABLE V
RELIABILITY AND CALIBRATION RESULTS OF THE PROPOSED MODEL.

Metric	Value
MCC	0.9959
Cohen's Kappa	0.9959
Macro Sensitivity	0.9968
Macro Specificity	0.9992
Brier Score	0.0064
ECE	0.0032
Accuracy (95% CI)	[0.9902, 1.0000]
Macro F1 (95% CI)	[0.9898, 1.0000]

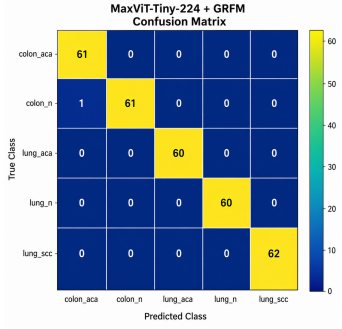


Fig. 4. Confusion matrix of the proposed MaxViT-Tiny-GRFM model on the 305-image test set.

D. Confusion Matrix and Class-Wise Performance

As shown in the confusion matrix in Figure 4, the model correctly classified every sample in four of the five classes. Specifically, it correctly identified all 61 colon adenocarcinoma images, 60 lung adenocarcinoma images, 60 normal lung images, and 62 lung squamous cell carcinoma images. Of the 62 normal colon images, 61 were correctly classified, while one was misclassified as colon adenocarcinoma. Thus, all lung tissue classes were separated without error, and the only confusion occurred between normal and malignant colon tissue.

E. RPR-AUC Analysis

The precision–recall performance of the proposed MaxViT-Tiny-GRFM model is presented in Figure 5. The model achieved excellent macro-averaged and weighted-average PR-AUC performance across the five histopathological classes. In addition, each individual class achieved a PR-AUC of 1.0000, indicating perfect ranking-based discrimination between positive and negative samples for every class.

The near-perfect precision–recall performance indicates that the proposed framework produces highly accurate class-probability rankings and effectively captures class-specific visual patterns. These results further suggest that the GRFM-enhanced feature representation improves the discriminative capacity of the model.

F. Cross-Validation and Seed-Stability Analysis

To assess the robustness and stability of the proposed MaxViT-Tiny-GRFM model, three-fold cross-validation and multi-seed experiments were conducted. As summarized in Table VI, the three-fold cross-validation produced a mean accuracy of 0.9979 ± 0.0007 and a mean macro F1-score of 0.9698 ± 0.0016 , with only small variations across the folds. These results indicate that the model performs consistently across different training partitions. Moreover, the three-seed evaluation presented in Table VII achieved a mean

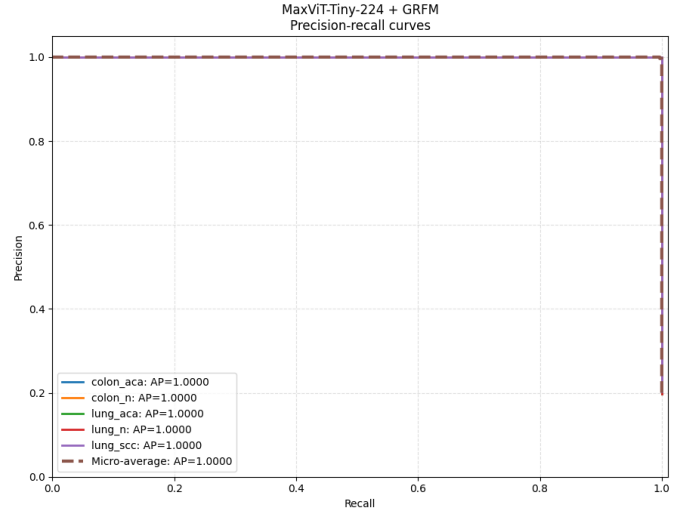


Fig. 5. Class-wise precision recall curves of the proposed MaxViT-Tiny-GRFM model.

TABLE VI
THREE-FOLD CROSS-VALIDATION RESULTS.

Fold	Accuracy	Macro F1
1	0.9971	0.9971
2	0.9983	0.9983
3	0.9983	0.9983
Mean $\pm$ SD	0.9979 $\pm$ 0.0007	0.9979 $\pm$ 0.0007

TABLE VII
THREE-SEED STABILITY RESULTS.

Seed	Accuracy	Macro F1
42	1.0000	1.0000
123	1.0000	1.0000
2024	0.9902	0.9902
Mean $\pm$ SD	0.9967 $\pm$ 0.0057	0.9967 $\pm$ 0.0057

test accuracy of 0.9967 ± 0.0057 and a mean macro F1-score of 0.9967 ± 0.0057 . Although the run with seed 2024 showed a slight decrease in performance, the overall results indicate that the proposed model is stable under different random initializations.

G. Robustness Under Image Corruptions

The robustness of the proposed MaxViT-Tiny-GRFM model under different image corruptions is summarized in Table VIII. The model achieved a mean accuracy greater than 0.98 and a worst-case accuracy greater than 0.94 under darkening, low-contrast, and occlusion conditions. In contrast, noise- and blur-based corruptions caused greater performance degradation. The lowest mean accuracy was observed under speckle noise at 0.8505, whereas the lowest worst-case accuracy was obtained under Gaussian blur at 0.6393. Overall, the proposed model is relatively robust to intensity changes and partial occlusion but is more sensitive to severe degradations that obscure fine histopathological structures.

H. Computational Efficiency and Resource Requirements

The proposed model contained 31.46 million parameters and required approximately 5.33 GMACs, equivalent to 10.67 GFLOPs,

TABLE VIII
ROBUSTNESS UNDER IMAGE CORRUPTIONS.

Corruption	Mean Acc.	Worst Acc.
Gaussian noise	0.8715	0.7639
Salt-and-pepper noise	0.9370	0.8361
Speckle noise	0.8505	0.7738
Gaussian blur	0.8708	0.6393
Motion blur	0.8977	0.7508
Darkening	0.9954	0.9902
Brightening	0.9161	0.7934
Reduced contrast	0.9836	0.9410
JPEG compression	0.9384	0.8000
Occlusion	0.9961	0.9934

TABLE IX
COMPUTATIONAL CHARACTERISTICS OF MAXViT-TINY-GRFM.

Measure	Value
Trainable parameters	31,458,381
Estimated parameter size	120.29 MB
MACs	5.33 GMAC
Estimated FLOPs	10.67 GFLOPs
Training time	35.45 min
Batch-1 model latency	33.51 ms
Batch-16 per-image latency	3.85 ms
End-to-end per-image latency	4.24 ms
End-to-end throughput	235.89 images/s

per forward pass as shown in table IX. Its estimated parameter-storage size was 120.29 MB. Training was completed over 23 epochs with early stopping and required approximately 35.45 minutes in a CUDA-enabled Kaggle environment, corresponding to an average epoch duration of approximately 92 seconds.

For a batch size of one, model-only inference required a mean latency of 33.51 ms, corresponding to approximately 29.85 images per second. With a batch size of 16, the mean per-image latency decreased to 3.85 ms, while throughput increased to approximately 259.66 images per second. End-to-end processing of all 305 test images required 1.293 seconds, corresponding to 4.24 ms per image and a throughput of 235.89 images per second. These results indicate that the model is computationally feasible for high-throughput offline analysis. However, its parameter count and GPU-memory requirements are greater than those of extremely lightweight architectures intended for mobile deployment.

I. GRFM Ablation Analysis

An ablation study was conducted to evaluate the contribution of the proposed Gated Residual Feature Modulation (GRFM) module. Without GRFM, the original MaxViT-Tiny backbone achieved an accuracy of 0.9738 and a macro F1-score of 0.9736. Incorporating GRFM increased both accuracy and macro F1-score to 0.9967, corresponding to improvements of approximately 2.29% and 2.31%, respectively. The proposed model contained 31,458,381 trainable parameters, whereas the original MaxViT-Tiny backbone contained 30,406,093 trainable parameters, representing an increase of approximately 3.46%. Thus, the modest increase in model complexity produced a substantial improvement in classification performance, indicating that GRFM effectively enhances the discriminative capacity of the extracted feature representations.

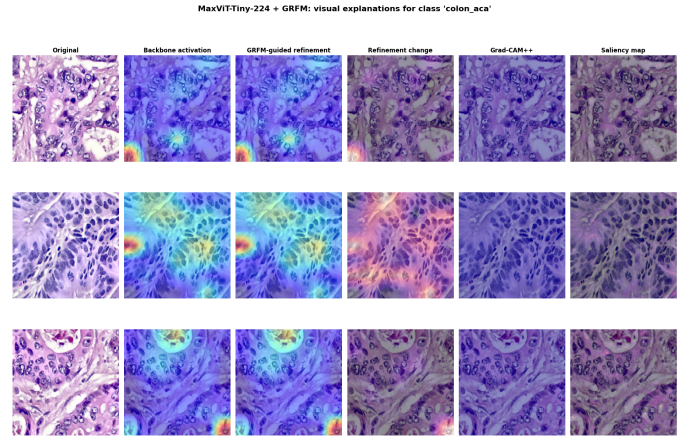


Fig. 6. Visual explanations generated by the proposed MaxViT-Tiny-GRFM model for three correctly classified colon adenocarcinoma (`colon_aca`) images. From left to right, each row presents the original histopathology image, backbone activation, GRFM-guided feature refinement, refinement-change map, Grad-CAM++ explanation, and gradient-based saliency map.

J. Proposed Model Explainability

Visual explanation techniques were used to identify the histopathological regions that contributed to the predictions of the proposed model and to better understand its decision-making process. Figure 6 presents visual explanations generated by the MaxViT-Tiny-GRFM model for three correctly classified samples from the `colon_aca` class.

The backbone activation maps broadly cover the discriminative tissue regions, whereas the GRFM-guided maps place greater emphasis on cellular clusters and glandular structures. The refinement-change maps show that GRFM selectively amplifies or suppresses features without erasing the backbone representation.

The Grad-CAM++ and saliency maps consistently highlight densely stained nuclei, irregular epithelial-cell organization, and glandular boundaries, as shown in Figure 6. Agreement among the different visualization methods suggests that GRFM preserves the discriminative features learned by MaxViT-Tiny while improving their contribution to the final prediction. These visualizations are not pathological annotations; rather, they provide qualitative evidence of model interpretability.

K. Comparison with Existing Studies

Table X compares the proposed framework with four recent methods for lung and colon histopathology classification. The selected studies report very high accuracies, ranging from 99.44% to 99.88%, suggesting that LC25000 has become a highly saturated benchmark. The studies by Ochoa-Ornelas and AlShehri focused primarily on architectural performance and achieved slightly higher raw accuracy than the proposed model. In contrast, the proposed MaxViT-Tiny-GRFM framework achieves 99.67% accuracy while incorporating duplicate-aware dataset preparation, calibration assessment, confidence-interval estimation, cross-validation, multi-seed analysis, corruption testing, computational profiling, ablation studies, and explainability. It also improves feature discrimination through a single-backbone gated residual refinement strategy rather than a multi-backbone ensemble. Therefore, the added value of the proposed framework lies not only in its classification accuracy but also in the breadth and transparency of its evaluation.

TABLE X
COMPARISON OF THE PROPOSED FRAMEWORK WITH EXISTING LUNG AND COLON HISTOPATHOLOGY CLASSIFICATION STUDIES.

Study	Core Method	Accuracy	Evaluation Highlights
Vanitha <i>et al.</i> [6]	Xception–MobileNet ensemble with hyperparameter tuning and Grad-CAM	99.44%	Combined complementary deep features with visual explanation; however, the evaluation remained primarily benchmark-centered.
Alotaibi <i>et al.</i> [7]	Channel-attention ResNet50, Tuna Swarm Optimization, and ensemble classifiers	99.60%	Reported multiple metrics under 80:20 and 70:30 data splits, but did not provide independent external validation.
Ochoa-Ornelas [12]	Cross-gated EfficientNetV2-B0 and MobileNetV3-Small dual-branch network	99.84%	Achieved a strong accuracy–efficiency trade-off through lightweight cross-branch attention and complementary feature fusion.
AlShehri [13]	DenseNet121 and MobileNetV3-Large with squeeze-and-excitation and multi-head self-attention	99.88%	Reported the highest accuracy among the selected studies using a dual-path attention-based architecture.
Proposed MaxViT-Tiny-GRFM	MaxViT-Tiny with gated residual feature modulation	99.67%	Used duplicate-aware data preparation and provided ablation, calibration, confidence intervals, cross-validation, seed stability, corruption robustness, efficiency profiling, and multi-level explainability.

The proposed model is not the most accurate of the selected studies, but it does have a 99.67% accuracy. It does more than just accuracy-centred comparisons, removing images that are almost identical before splitting up and jointly assessing calibration, statistical stability, corruption robustness, computational cost, ablation and explainability. Hence, the key benefit of MaxViT-Tiny-GRFM is to provide high discriminative ability together with a more transparent and rigorous protocol for validation. External multi-center evaluation, however, will still be needed to ensure clinical generalizability.

V. CONCLUSION, LIMITATIONS, AND FUTURE WORK

This study proposes a GRFM-based MaxViT-Tiny framework for the classification of lung and colon histopathology images into five categories. To minimize evaluation bias caused by exact and near-duplicate images, duplicate samples were removed before data splitting. The proposed Gated Residual Feature Modulation module was designed to enhance transformer features through adaptive gating while preserving residual information. On the unseen test set, the model achieved 99.67% accuracy, 99.68% macro precision, 99.68% macro recall, and 99.67% macro F1-score. It also achieved MCC and Cohen’s kappa values of 0.9959, while the low Brier score and ECE indicated well-calibrated probability estimates. The effectiveness of the framework was further examined through cross-validation, multi-seed testing, corruption analysis, computational profiling, ablation studies, and visual explanations. The ablation study showed that adding GRFM improved accuracy from 97.38% to 99.67% with only a slight increase in the number of trainable parameters. Nevertheless, the study has several limitations. Evaluation was conducted on a subset of LC25000, and external multicenter clinical datasets were not included. The model also showed reduced robustness under severe blur and noise and required more computational resources than mobile-oriented networks. Future work will focus on external validation, stain normalization, domain adaptation, corruption-aware training, model compression, and lightweight deployment. Expert evaluation of the Grad-CAM++ explanations will also be investigated to improve clinical interpretability and utility.

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